

Predicting Movie Revenues Using Machine Learning Algorithms

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Abstract

This research aims to develop robust machine learning models for predicting movie revenues and profitability, a fundamental task for the cinema industry. This was achieved through a comprehensive methodology that involved integrating and processing diverse, feature-rich data sources from platforms like Kaggle and IMDb, in addition to data on actors and Academy Awards. A wide range of machine learning models was applied, including Linear Regression, Decision Trees, Random Forest, XGBoost, and CatBoost. Experiments showed that the Random Forest model delivered effective performance in predicting movie revenues, while the CatBoost model excelled, achieving excellent accuracy in the task of classifying movie profitability (profitable/loss-making), along with successful handling of data challenges such as zero values. The achieved results confirm the contribution of this research to the field, as the models demonstrate competitive and robust performance compared to previous works, providing a scientific foundation for supporting strategic decision-making in the cinema industry.

I. Introduction

The film industry is one of the major economic sectors globally, characterized by significant investment risks. Amid increasing competition and rising production costs, understanding the factors that contribute to a film's commercial success has become critically important for producers, investors, and decision-makers. Profitability heavily depends on the ability to predict a film's potential revenue before its release or in its early stages, allowing for sound decisions regarding production

aspects, marketing, and budget allocation.

Furthermore, the selection of the cast is known to play a pivotal role in attracting audiences and determining the artistic and commercial value of a film, making the process of selecting the right actors a complex challenge requiring precise analysis.

This study aims to address these challenges by developing a comprehensive machine learning model and implementing integrated data mining, focusing on predicting movie revenues and providing an intelligent actor recommendation system. Unlike traditional studies that may rely on limited datasets or superficial features, this project builds on constructing a unique and expanded dataset. Multi-source data was integrated, including basic movie information, user ratings, and detailed actor data from databases like IMDb, including actor ratings, in addition to advanced engineered features extracted from budget and initial revenue data. This rich set of features allows for deeper exploration of the complex relationships between various production factors and a film's financial performance.

The key contributions of this paper are as follows: Building and assembling a non-traditional, comprehensive dataset for movie revenues, including rich, specially engineered features about actors and movie performance.

Conducting an in-depth exploratory data analysis to uncover hidden patterns and potential issues like missing values and imbalanced distributions.

Applying advanced feature engineering techniques to create new metrics with significant impact on revenue prediction.

Applying and comparing several machine learning algorithms for predicting movie revenues, with fine-

tuning of hyperparameters and performance evaluation using multiple metrics.

Developing a data-driven actor recommendation system aimed at assisting filmmakers in selecting the appropriate cast.

Providing a detailed comparative analysis of model results against baselines, discussing model performance based on their complexity and training time, and identifying the most influential factors.

The rest of this paper is organized as follows:

Section II discusses related work. Section III reviews the dataset description and exploratory data analysis. Section IV presents the data preprocessing steps and feature engineering. Section V details the machine learning models and experimental methodologies. Section VI presents the results and performance analysis. Section VII discusses the comparison of the work with baseline or state-of-the-art techniques. Finally, Section VIII presents the conclusion and future work.

II. Related work

The film industry is a field of major economic and cultural importance, driving the urgent need to develop accurate models for predicting movie revenues and success. Numerous studies have addressed this problem using various techniques and different datasets.

Some studies focused on predicting revenue based on pre-release factors. For example, [1] Xu (2024) predicted box office revenue using features like budget, popularity, genre, and runtime, achieving an accuracy of up to 73% using a Random Forest model. Although this study confirmed the importance of budget and popularity, it pointed to the need for improving data quality and quantity. Similarly, the study by Udandarao and Gupta [2] (2024) explored a wide range of machine learning algorithms for regression, including ensemble models, and concluded with strong performance (R-squared - 0.86, RMSE - 8.26 million dollars) on a larger dataset. This study explicitly stated that considering factors like the director, writer, and actors represents an area for future work.

On the other hand, other studies addressed different aspects of movie success or used deeper techniques.

Agarwal et al. (2021) [3] conducted a comprehensive study predicting "movie success rate" as a classification problem based on "Metascore" from IMDb, using a large dataset and detailed user rating features, showing neural networks performed best (86% accuracy). In the context of deep learning, Zheng (2024) [4] predicted box office revenue using models like Bidirectional LSTM and XGBoost, emphasizing the "significant impact of star actors and directors" on revenue.

Despite the progress made by these studies, our current project stands out by complementing these efforts and addressing key gaps. While previous studies focused on predicting revenue [1, 2, 4] or classifying success [3], they often lacked the integration of deep features about actors or a quantitative, detailed analysis of their impact in prediction models. Furthermore, these studies did not address the aspect of actor recommendation systems, which is of significant practical importance to filmmakers. Therefore, this work is distinguished by compiling and building a unique and comprehensive dataset that includes advanced, specially engineered metrics for evaluating actors (such as their average ratings and additional information about them), in addition to integrating the revenue prediction task with the development of an actor recommendation system, providing a more integrated analytical and recommendation tool for the film industry sector.

III. Dataset Description and Exploratory Analysis

1. Dataset collection

To ensure the comprehensiveness and accuracy of predicting movie revenues and the actor recommendation system, a unique and expanded dataset was built from multiple sources. The data collection process centered around integrating public movie datasets with detailed actor data to create a rich database capable of capturing the complexities of the cinema industry and success metrics.

The primary data source was the "Kaggle daily update dataset" [5], which provides basic information about movies such as budget, revenue, popularity, genres, and release dates. To complement this data, additional information about movies and actors was extracted from the IMDb database [6]. This data was carefully filtered to ensure its suitability for the project's goals; documentary films and TV series were excluded, and the focus was narrowed to English-language films to ensure dataset homogeneity and focus.

The most distinctive part of the data collection and engineering process was the creation of a custom actor dataset and the extraction of movie ratings. By analyzing the extracted data from the IMDB dataset, three unique evaluation metrics were developed for each actor, added as new features to the main dataset:

1. **Average Actor Rating:** Represents the average IMDB rating of all movies the actor has participated in throughout their career.
2. **Bayesian Average Actor Rating:** The Bayesian average was used to smooth actor ratings and provide a more stable estimate. This method weights the actor's average rating with the global average, giving more weight to actors with a higher number of movies. This approach ensures reliable and fair ratings, especially for actors with fewer records, by pulling their ratings towards the global average.
3. **Penalized Actor Rating:** A custom penalty system was developed to adjust the actor's average rating based on the total number of votes their movies received. A discount factor (denominator) is applied to the average rating, where the discount becomes larger the lower the total number of votes for the actor, giving less weight to actors who have a negligible number of votes in their films.

These new actor metrics were merged with the main movie dataset. This process resulted in adding rich features that significantly enhance the models' ability to predict revenue based on the cast.

The final dataset, after merging these sources and preprocessing, consists of 37,093 records and includes 44 features. This set provides a robust and

comprehensive database containing demographic information about the films, financial data, audience ratings, and specially engineered metrics on actor performance, making it perfectly suitable for revenue prediction tasks and creating a recommendation system.

2. Exploratory Data Analysis

• Distribution of Key Numerical Features:

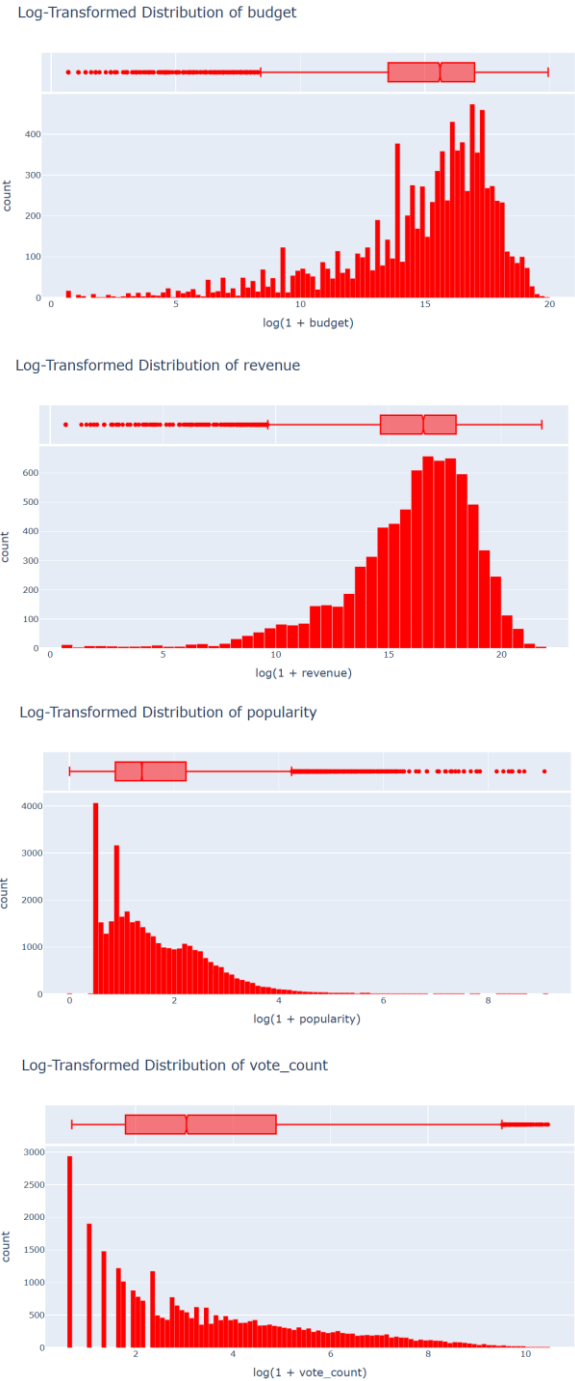


Figure 1 Distribution of Numerical Features

Exploratory analysis of the distributions of numerical features revealed important

characteristics requiring data preprocessing. Features like budget, revenue, popularity, and vote_count showed a strong positive skew. This indicates that the majority of films fall within low-value ranges for these features, with a very small number of films having extremely high values, such as films with massive revenues or large budgets.

A logarithmic transformation was applied to the highly skewed features, which helped bring their distributions closer to normal and facilitated the detection of underlying patterns.

• **Analysis of the Overall Trend in Movie Production Over the Years:**

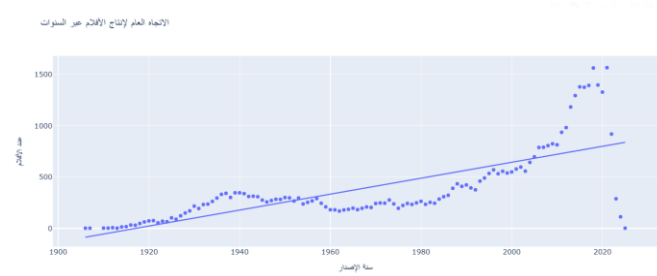


figure 2 Overall Trend in Movie Production Over the Years

Time series analysis of the number of movies produced annually showed a clear upward long-term trend, especially since the early 1990s. The period from 2000 onwards saw rapid jumps in production, which can be linked to technological development and the proliferation of streaming platforms. Growth was gradual and slow between 1900 and 1950 (the early stages of the cinema industry), and a decrease in production was observed between 1940-1980, perhaps due to the effects of World War II. Towards the end of the 1990s and the beginning of the millennium, a boom occurred. However, a sharp decrease in the number of movies produced was observed after 2020, showing the impact of the COVID-19 pandemic and the postponement or cancellation of many releases, in addition to the potential incompleteness of data for recent years and increased demand for online streaming platforms.

• **Analysis of the Most Common Genres and the Most Profitable Genres:**

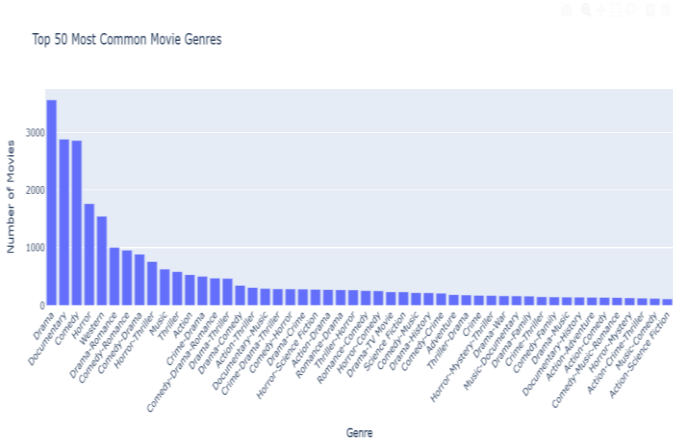


figure 3 Top 50 Most Frequent Movie Genres

The analysis revealed that the distribution of movie production by genre is highly imbalanced, with a clear focus on Drama, Comedy, and Documentary films as the most produced genres within the data.

When considering net profitability, it was found that genres combining Action and Thriller (Action-Thriller), Comedy with Romance (Comedy-Romance), or Comedy, Drama, and Romance (Comedy-Drama-Romance), in addition to Action-Drama and Romance-Comedy, achieve the highest net profits. These results indicate that combining elements from these genres often attracts wider audiences and achieves excellent returns on investment. It was also noted that genres with relatively low budgets, such as Horror and Documentary, can be very profitable compared to their cost. Meanwhile, the least profitable (or loss-making) genres were concentrated in categories like Action, Music, and Science Fiction, indicating the need for careful scrutiny of projects within these categories. It is concluded that there is a clear relationship between smart investment in medium budgets and achieving large profits, and that very high investment can achieve massive profits in specific genres like Action-Thriller.

• **Analysis of the Top 20 Production Companies by Number of Movies Produced and Their Genres:**

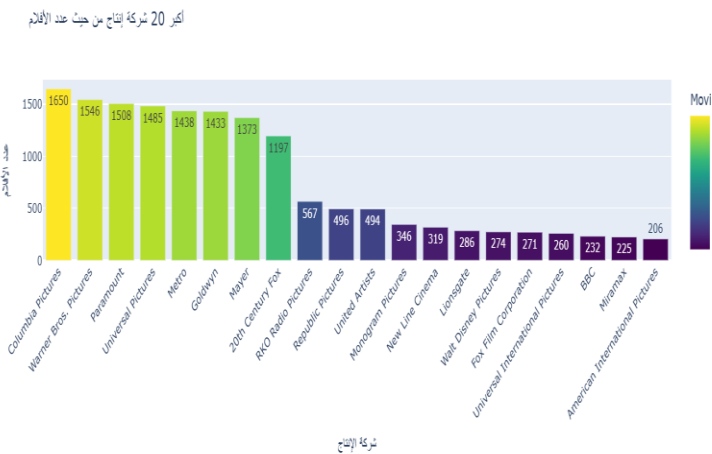


figure 4 Top 20 Companies by Number of Movies

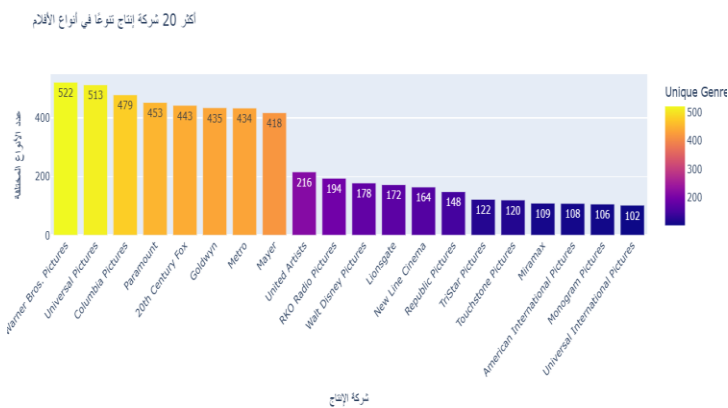


figure 5 Top 20 Companies by Diversity of Movie Genres

It was found that the companies that produce the most movies are also the most diverse in terms of the types of movies they produce, such as Warner Bros. Pictures, Universal Pictures, Columbia Pictures, Paramount, and 20th Century Fox, which produce hundreds of different types of movies. This suggests that major companies seek to balance "mass production" and "genre diversity" to target different audiences and increase profits. However, there is no direct and inevitable relationship between diversity and production volume; some specialized companies may prefer to focus on specific genres to build a strong identity in a particular market, which can enhance their profit opportunities in that field.

• Distribution of Movie Ratings:

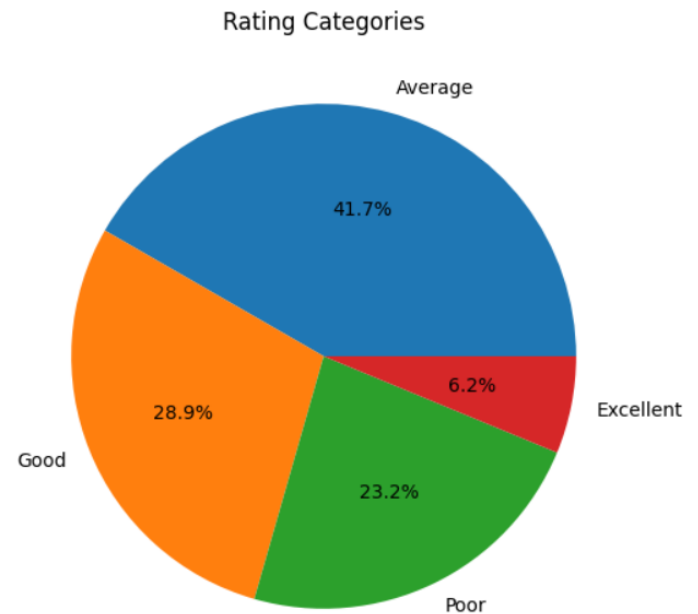


figure 6 Distribution of Movie Ratings

Analysis of the distribution of movie ratings showed that the majority of films fall into the "Average" category, followed by the "Good" category. While "Excellent" films constitute a small percentage, and "Weak" films constitute a non-negligible percentage. This distribution indicates that most films achieve acceptable to good ratings, with a very small minority of highly distinguished films.

• Relationship Between the Number of an Actor's Films and the Highest Rating of a Film They Participated In:

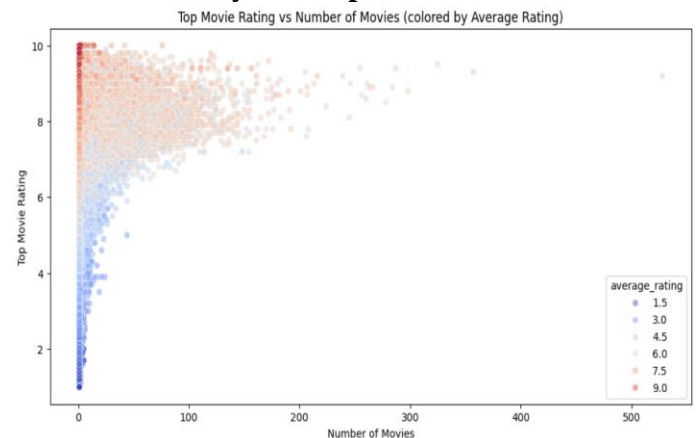


figure 7 : Ratings vs. Number of Films

The analysis revealed an interesting relationship: actors with a small number of films often show a very high "highest movie rating" (more than 8), and their overall average rating tends to be high. As the

number of films an actor participates in increases, the distribution tends to become more homogeneous; while films with high ratings continue to exist, the presence of actors with an extremely high "highest movie rating" (9-10) becomes non-existent, and high-rated films (above 8) become less dense. In other words, it is easier for an actor with a small number of films to have one film with a very high rating compared to an actor who has participated in a large number of films, where their overall average rating becomes more stable and may reduce the likelihood of them participating in a film with an exceptionally high rating within their broad career.

- **Correlation Matrix:**

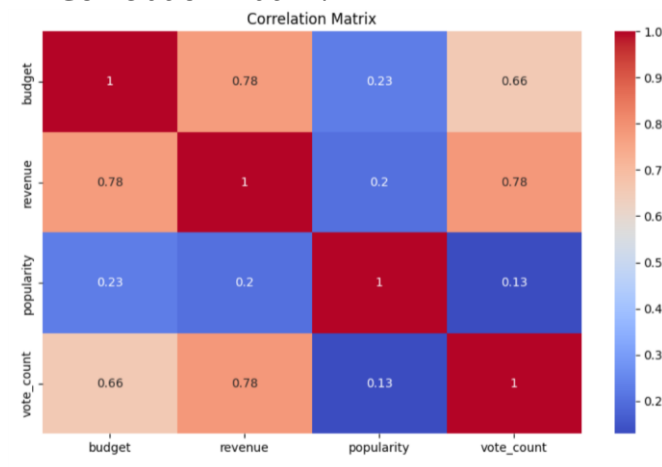


figure 8 Correlation Matrix

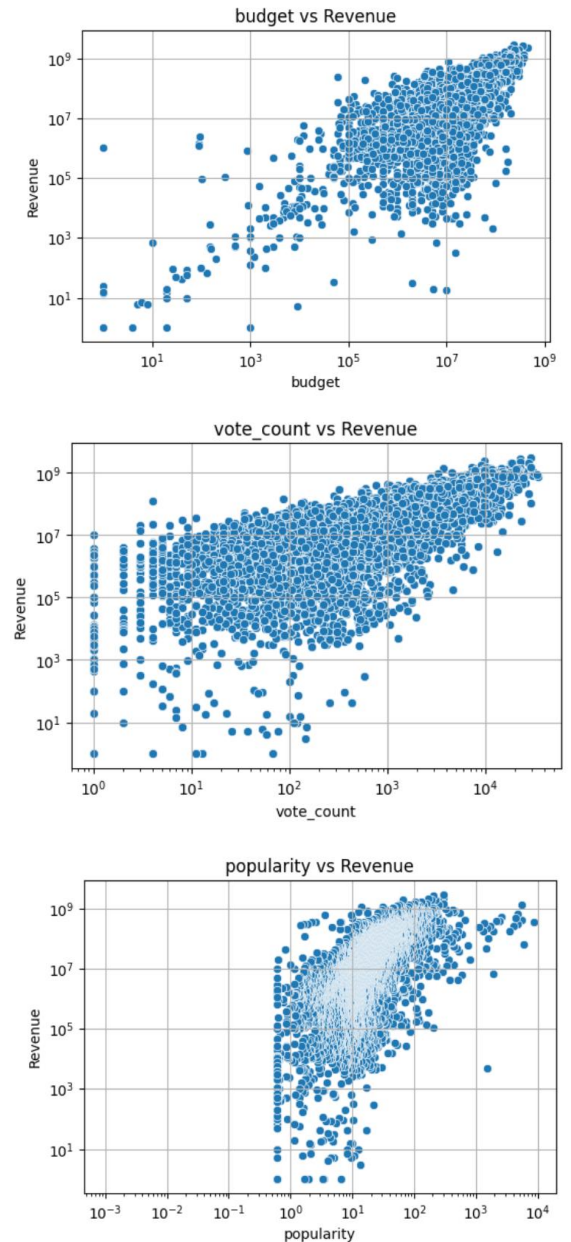


figure 9 Correlations More Clearly

The correlation matrix showed strong and clear relationships between financial features and rating features:

- Very strong relationship between Budget and Revenue: The positive correlation coefficient reached 0.78, indicating that increased budget is significantly associated with increased revenue achieved.
- Very strong relationship between Revenue and Number of Votes (vote_count): The positive correlation coefficient reached 0.78, indicating that films with high revenue also tend to receive a large number of ratings from the audience.

- Good relationship between Budget and Number of Votes: The positive correlation coefficient reached 0.66, indicating that films with large budgets often attract greater audience attention and receive more votes, which is natural since a large part of the budget goes towards marketing.
- Weak correlations with Popularity: Contrary to expectations, it showed weak correlation with Budget (0.23), Revenue (0.20), and Number of Votes (0.13). This indicates that the "popularity" metric in this data may depend on other factors or has a different definition that is not strongly reflected in these financial variables and direct audience ratings.

It is clear from the exploratory analysis that Budget and Number of Votes are two key factors with a significant impact and strong correlation with Revenue in this dataset. Also, engineering the actor rating features added new dimensions to the analysis, demonstrating the importance of these features in understanding the dynamics of movie success.

IV. Data Processing and Feature Engineering

The data processing and feature engineering phase is crucial to ensure data quality, improve its performance in machine learning models, and extract the maximum possible value from it. Given the complex nature of the dataset and the integration of multiple sources, this stage involved intensive processes of cleaning, transformation, and creation of new features. The explanation will be divided according to the main components of the dataset to provide a clear view of the steps taken.

1. Processing the Basic Movie Dataset

The original movie dataset from Kaggle formed the starting point and underwent the following steps:

a .Initial Data Processing and Cleaning

1. Filtering movies by language: Only films whose original language was English were retained.

2. Deleting unnecessary columns: Several columns that were not directly relevant to prediction or recommendation tasks, or contained data not easily usable, were removed. These columns included: keywords, poster_path, backdrop_path, and recommendations.
3. Handling missing values and duplicate rows:
 - Deleting rows with missing and duplicate values: Due to a high percentage of missing values in columns sensitive and important for analysis, the decision was made to delete all rows containing any empty values to rely only on complete data records. Any duplicate rows were deleted, keeping the first occurrence.
 - Deleting duplicates based on title and release date: To prevent duplicate movies that might appear multiple times with slightly different records but the same title and release date, duplicates were removed.
 - Result: After these steps, the number of remaining films in the dataset stabilized at 48,654 films.
4. Handling zero runtime values and outliers (runtime):

Films whose runtime was equal to zero were excluded.

An additional check was performed for films with very short runtimes (less than 15 minutes) and very long runtimes (more than 400 minutes). After verifying that they did not belong to specific genres that might justify these values (like very short films or documentaries), the decision was made to filter the dataset to retain only films with runtimes between 15 and 400 minutes. This ensures a focus on traditional films and removes outliers that could negatively affect modeling.

b. Feature engineering

Several new features were derived from existing columns to enrich the dataset and increase the models' ability to learn and capture relationships:

1. Profit feature (profit): The net profit for each film was calculated by subtracting the budget value from the revenue.
2. Detailing the release date (release_date): The release_date column was split into its temporal components to create new features:
 - release_year: Year of film release.
 - release_month: Month of film release.
 - release_day: Day of film release.

This helps detect seasonal patterns or trends related to the year of release.

3. Number of unique actors (unique_actors_count): The number of unique actors participating in each film was extracted from the credits column, which contains actor names.
4. Sentiment analysis of the tagline (tagline_sentiment): Sentiment analysis was applied to movie taglines to create a new feature reflecting the emotional tone of the phrase (positive, negative, or neutral). This feature aims to provide the machine learning model with additional information that could be an important indicator of the film's appeal to the audience and thus its potential revenue. For example, a positive tagline might enhance viewer interest and increase revenue.
5. Movie era classification (movie_era_classification): A new categorical feature was created that classifies the film into different time eras based on its release year, to capture the effect of historical and cultural changes in the film industry:
 - 'Classic': Before 1980
 - 'Old_School': From 1980 to before 2000
 - 'Modern': From 2000 to before 2015
 - 'New_Era': From 2015 onwards
6. Budget category classification (budget_category): Budget was converted into categories to facilitate analysis and understanding, and perhaps to capture non-linear relationships:
 - 'low_budget': Less than 1 million dollars

- 'medium_budget': From 1 million to less than 20 million dollars
- 'high_budget': From 20 million to less than 100 million dollars
- 'blockbuster': 100 million dollars or more

2. Processing the Full IMDb Dataset

The IMDb dataset, used to provide additional information about movies and actors, underwent specific cleaning and feature engineering processes to ensure its quality and suitability for the task::

a. Initial Data Processing and Cleaning

1. Handling missing values: Missing values in the genres column were handled specifically by replacing them with the value "Unknown", to ensure the completion of this important column. Other columns were checked for missing values and handled within a comprehensive strategy.
2. Deleting unnecessary columns: Unnecessary columns such as isAdult and endYear were removed because the focus was on general films.
3. Filtering for movies only: To ensure the dataset contained only traditional films, records were filtered to retain those where titleType equals 'movie'.
4. Deleting unwanted genres: Films belonging to certain genres irrelevant to predicting box office revenue for traditional commercial films were excluded. These genres included: 'Reality-TV', 'News', 'Documentary', 'Biography', 'Adult', 'Game-Show', 'Short', 'Sport', 'music', and 'Talk-Show'.
5. Handling illogical years: The startYear column was checked to identify any illogical values (e.g., years before 1900 or exceeding 2025). These records were identified to ensure all films are within a reasonable time range.
6. Handling abnormal runtime values (runtimeMinutes):
 - The presence of extreme and illogical values was noted, ranging from one

minute to approximately 59,460 minutes (about 41 days).

- To ensure a focus on traditional films with reasonable runtimes, the dataset was filtered to retain only films with runtimes between 30 and 300 minutes.

7. Adding average movie rating: The average rating (averageRating) for each film from the ratings dataset was merged into the base IMDB dataset using the unique film identifier (tconst) as a linking column.

b. Feature engineering

New features were derived from existing columns in the IMDB dataset to enhance its analytical capability:

1. Runtime classification (runtime): The numerical film runtime was converted into descriptive categories to facilitate analysis and modeling:
 - 'short': Less than 75 minutes.
 - 'standard': From 75 to 120 minutes.
 - 'long': More than 120 minutes.
2. Movie age (movie_age): The age of the film (up to the current year 2025) was calculated by subtracting the release year (startYear) from 2025.
3. Rating category (rating_bucket): A new categorical feature was created based on the film's average rating to classify its quality:
 - 'Excellent': 8 or higher.
 - 'Good': From 6.5 to less than 8.
 - 'Average': From 5 to less than 6.5.
 - 'Poor': Less than 5.

3. Processing Actor Data

a. Extracting and Aggregating Actor Information

1. Filtering and standardizing actor information:

- The table containing actor roles in films was initially filtered to retain only records pertaining to actors.
- This table was then merged with the names table (using the person identifier column nconst) to obtain the primary names of actors associated with each film role.

2. Aggregating actors per film:

- The names of all actors participating in each film were aggregated into a single new column. This was done by grouping by the film identifier (tconst) and removing any duplicate names from the IMDB data, adding a column for the best film.

3. Cleaning the merged actor data:

- The actors_in_movie column was converted to string type, and missing values were filled.
- Rows that did not contain actor information were filtered out.

b. Feature engineering

To more accurately represent the quality of actor performance, three different rating metrics were developed for each actor, in addition to information about their best film. This process first required merging the number of votes (vote_count) from the IMDB movie dataset, as it is crucial for the penalty system.

1. Simple average of actor ratings (average_rating):

- This represents the direct arithmetic mean of all the ratings of the films the actor participated in.
- This metric provides a quick overview of the actor's performance based on all their work.

2. Bayesian average of actor ratings (bayesian_avg_rating):

- This metric aims to provide a more stable and reliable rating for actors, especially those with few films. It is calculated as a weighted average between the global

average of all actor ratings (global_avg) and the simple average of the individual actor.

- The formula applied is: $(C * \text{global_avg} + \text{sum_ratings}) / (C + \text{num_movies})$
- Where C is the confidence coefficient (set to 5, a tunable value), global_avg is the global average rating of all films, sum_ratings is the sum of the ratings of the films the actor participated in, and num_movies is the number of films the actor participated in.
- This approach pulls the ratings of actors with a small number of films towards the global average, preventing overestimation or underestimation of their value based on a small sample.

3. Actor rating with penalty for low number of votes:

- This system aims to penalize (reduce) the actor's average rating if the total number of votes their films received is low, indicating low exposure to audience rating.
- The total sum of votes (total_votes) for all films the actor participated in is calculated.
- A discount factor (divisor) is applied to the actor's simple average (average_rating) based on total_votes according to the following thresholds (determined based on vote distribution analysis):
 - If less than 10: divisor = 4
 - If between 10 and 30: divisor = 3
 - If between 31 and 100: divisor = 2
 - If greater than 100: No discount (divisor = 1)
- $\text{penalty_avg_rating} = \text{average_rating} / \text{divisor}$
- This ensures that actors whose films do not receive many ratings appear with a more conservative rating.

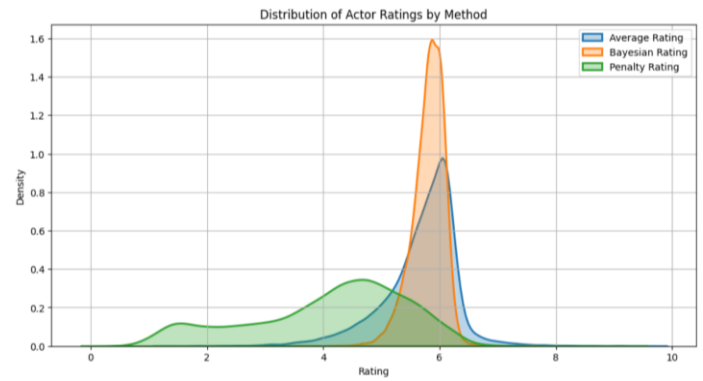


figure 10 Comparison of Rating Distributions Using the Three Methods

4. Information on the actor's best film:

- For each actor, the name of their highest-rated film (top_movie_name) and the rating of this film (top_movie_rating) were also extracted.

Results:

These processes resulted in a separate actor dataset table, which was later merged with the movie data to form the cast rating feature.

- Through descriptive statistics, it was noted that the Bayesian average had a much lower standard deviation compared to the simple average, confirming its effectiveness in smoothing ratings.
- In contrast, the penalty rating showed a lower average and greater dispersion, reflecting the application of discounts to actors with few ratings.

4. Processing Oscar Awards Data and Feature Engineering

To add a dimension related to prestigious awards, Oscar awards data [7] was integrated into the analysis, with processing and feature engineering operations to accurately link it to movie and actor data.

a. Initial Data Processing and Cleaning

Standardizing titles for matching: The number of films that matched between the two datasets was determined after performing transformations to

standardize names (removing spaces and converting to lowercase), which provided a basis for linking the Oscar data. The count was 3097.

b. Feature engineering

Three main features related to Oscar awards were derived from the raw data:

1. Film's Oscar status (movie_oscar):

- A column was converted to a numerical value: 2 if the film was a winner, and 1 if it was only nominated.
- A map was then created linking each film to the number of Oscar statuses it achieved (nomination or win).
- This map was applied to the column of film titles. This feature reflects the level of Academy recognition for the film itself.

2. Crew's Oscar status (movie_credits_oscar):

- To assess the impact of Oscar-winning actors or directors in a film, the feature movie_credits_oscar was created.
- First, the names of individuals (actors, directors) in oscar_df were standardized by converting them to lowercase and removing spaces and non-alphanumeric characters.
- The numerical Oscar status for each person was determined.
- A map was created linking each standardized person name to the highest Oscar status they achieved (win or nomination).
- Then, for each film, the actor names were extracted from the credits column (after standardization and splitting), and the map was used to determine the number of Oscar statuses among all key individuals involved in the film.
- This feature aims to measure the "Oscar weight" that the crew brings to the film.

3. Total Oscar Awards for Production Companies (company_oscars):

- Using the same processing methods as before, the total number of Oscar instances (1 for nomination, 2 for win) was calculated for each production company across all the films it produced that appeared in the Oscar data.

- A function was created that aggregates these values for each film based on its associated production companies.
- This feature reflects the experience and general acclaim of production companies in winning or being nominated for Academy Awards, which may be an indicator of their quality and ability to produce successful films.

3. Processing Data for Top Actors (Movie Stars)

To create a feature that identifies the most famous and highest-earning "movie stars," data from two main sources [8] was analyzed and merged.

a. Description of Data Sources

1. Celebrity.csv:

- Contains general information about celebrities.
- Initial size: 9980 rows and 8 columns.

2. forbes_celebrity_100.csv:

- Contains the Forbes list of the highest-earning celebrities.
- Initial size: 1647 rows and 4 columns.

b. Cleaning Celebrity Data (Celebrity.csv)

1. Deleting unnecessary columns: The columns Unnamed, adult, gender, and id were removed as they were not needed for this analysis.
2. Filtering by acting field: Only records where the celebrity belonged to the acting field were retained.
3. Removing non-English names
4. Removing duplicates
5. Popularity analysis and selecting the top 150: After cleaning the data, a descriptive analysis of the popularity feature was performed. The data was then sorted in descending order by popularity, and the top 150 personalities were selected.
6. Final size: The cleaned celebrity dataset contained 5946 rows and 4 columns.

c. Cleaning Forbes Data (forbes_celebrity_100.csv)

1. Filtering the acting category: The data was filtered to retain only actors. This resulted in 176 actors and 107 actresses.
2. Handling duplicates: Done by extracting only the most recent appearance for each personality.
3. Sorting by income: The results were sorted in descending order by the Pay value to identify the highest-earning celebrities.
4. Final size: The cleaned Forbes dataset contained 524 rows and 4 columns.

d. Matching and Merging the Two Files

1. Standardizing names for matching: To facilitate accurate matching between the two datasets, the names of celebrities in both datasets were converted to a unified format (lowercase, without spaces or special characters).
2. Extracting the intersection: The common names between the two datasets were identified after the standardization process. The number of unique matching names was 73.
3. Filtering and merging: Only records containing these matching names were kept in both the celebrity and Forbes datasets. Afterwards, the two datasets were merged using the standardized Name column as the key.
4. Handling duplicates after merging: The most recent income record for each personality was kept after the merge, resulting in a final list of 73 unique personalities.
5. is_superstar feature: A feature (is_superstar) was added and its value was set to True for all records containing a star from the movie data to distinguish these personalities as "movie star" data points.

V. Machine Learning Models and Experiments

In this section, the models used, the experimental methodology, and the results achieved in the endeavor to predict movie revenue and profitability will be reviewed, highlighting the challenges and solutions adopted.

a. Data Preparation for Model Training

Before training the models, a fundamental step was taken to remove features that directly revealed the target variable (revenue or profit) or that degraded model performance. The following features were removed:

id, tconst, primaryTitle, originalTitle, data identifiers, normalized_title.

Metadata or temporal information that may not directly contribute as strong independent variables to prediction after feature engineering: title, tagline, release_date, startYear, runtimeMinutes, genres_y, status.

averageRating, movie_oscar

Features related to rating or Oscars.

The linguistic dataset was split into three parts: training data, validation data, and test data for an objective probability evaluation of model performance.

b. Target Variable Testing

A preliminary experiment was conducted to evaluate which of the two target variables (Profit or Revenue) provided a better tool for prediction. The results on the final test set showed the following:

R2	Best model	Target variable
0.6005	Random Forest	Earnings
0.7761	Random Forest	Revenue

Table 1: Selection of the Target Variable

Based on these results, the decision was made to rely on Revenue (Revenue) as the primary target variable due to its superior predictive performance.

c. Models Used and Initial Results

A set of medium-level machine learning models were trained and tested for predicting revenue. The initial results for revenue prediction (on training data):

Model	R ² Score
Linear Regression	0.778514
Decision Tree	0.686897
Bagging	0.791188
Random Forest	0.820138

XGBoost	0.801316
Gradient Boosting	0.781913
LightGBM	0.793551
CatBoost	0.812622

Table 2 models used

The Random Forest model showed a clear initial performance with an R^2 value of 0.82 on the training data and 0.77 on the test set used for revenue.

d. Challenges and Experiments

The revenue prediction process faced a major challenge represented by the large number of films with small revenues and budgets, which are unrealistic and negatively affect data distribution and model performance. To address this challenge, several experiments were conducted:

1. Application of Logarithmic Transformation and Clipping:

- Motivation: To address the non-normal distribution of revenue data and reduce the impact of outliers and skewness. A logarithmic transformation was applied to the target variable, with all values less than 1 set to zero before transformation.
- Evaluation Metrics: Model performance was evaluated using R^2 , Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

Model	R^2	MAE	RMSE
Linear Regression	-33588.1	3.58952e+08	1.42802e+10
Decision Tree	0.6433	1.0905e+07	4.6536e+07
Bagging	0.7499	7.70838e+06	3.89656e+07
Random Forest	0.775	7.25141e+06	3.69612e+07
XGBoost	0.7611	7.58782e+06	3.80872e+07
Gradient Boosting	0.6681	7.95966e+06	4.4891e+07
LightGBM	0.7581	7.3896e+06	3.83229e+07
CatBoost	0.7405	7.27423e+06	3.96886e+07

Table 3 Application of Logarithmic Transformation and Clipping

The Random Forest model remained the best in this experiment as well, achieving an R^2 of 0.775.

2. Conditional Regression after Binary Classification:

Motivation: To address the problem of zero values more directly, a two-step approach was used:

Step One: Build a binary classification model to predict whether the revenue would be zero or non-zero.

Step Two: If the revenue is non-zero, pass it to a separate regression model to predict the actual revenue value.

Results:

Model	R^2	MAE	RMSE
Linear Regression	-	230348138	2684052432
	19306.37	0	4
Random Forest	0.753	48206994	95985253
XGBoost	0.692	55335522	107153341
Gradient Boosting	0.654	53359829	113568278
LightGBM	0.766	47414590	93502160
CatBoost	0.739	48193353	98650595

Table 4 Application of Conditional Regression after Binary Classification

Conclusion: Despite this approach, it was noted that the model still struggled to accurately predict some values, indicating significant errors in some cases.

3. Binary Classification for Profit/Loss:

Motivation: Given the challenges in predicting the exact revenue value due to zero values, the problem was transformed into a classification question: Is the film profitable or loss-making? (If revenue is greater than the budget, it is profitable).

Handling Class Imbalance: It was noted that 80% of the films were "loss-making", representing a significant class imbalance. The weight parameter in the XGBoost model was used to balance the importance between classes (similar to resampling) to address this problem.

Evaluation Metrics: Classification metrics such as Accuracy, Recall, Precision, and F1-score were used.

Results (XGBoost):

	precision	recall	f1-score	support
0	0.93	0.96	0.94	6276
1	0.74	0.58	0.65	1143
accuracy			0.90	7419
macro avg	0.83	0.77	0.80	7419
weighted avg	0.90	0.90	0.90	7419

Table 5 XGBoost results

Confusion Matrix: $[[6044, 232], [479, 664]]$

Number of errors: 711 out of 7419

Results (CatBoost) - Best performance in classification:

	precision	recall	f1-score	support
0	0.93	0.97	0.95	6276
1	0.78	0.63	0.70	1143
accuracy			0.92	7419
macro avg	0.86	0.80	0.82	7419
weighted avg	0.91	0.92	0.91	7419

Table 6 CatBoost results

The CatBoost model showed the best performance in the binary classification task for Profit/Loss, achieving an overall accuracy of 92%.

VI. Results and Performance Analysis

In this project, the focus was on two main tasks:

1. Predicting Film Revenue (Revenue Prediction):

Best Model: Random Forest

Performance (without log transformation): $R^2 = 0.7761$ on the test set.

Performance (after log transformation): $R^2 = 0.775$, MAE = 7.25m, RMSE = 36.96m.

2. Classifying Film Profitability (Profit/Loss Classification):

Best Model: CatBoost

Performance: Accuracy = 0.92, F1-score = 0.70 (for the positive class – profitable films).

VII. Comparison of Work with Baseline or State-of-the-Art Techniques

a. Comparison with Referenced Studies

To assess the effectiveness and importance of the proposed model in this project, a comprehensive comparison was made with a set of previous research in the field of predicting movie revenue and success. This comparison aims to place the achieved results in the context of current knowledge and identify strengths and weaknesses compared to baseline and modern techniques and methods.

1. "A Comprehensive Study on Various Statistical Techniques for Prediction of Movie Success" (Agarwal et al.)

- ❖ Paper Goal: Predict movie success (classified as successful, failed, average) based on Metascore.
- ❖ Models and their Performance (Accuracy): Neural Network (ANN) was best with 86% accuracy, followed by Logistic Regression with 76% accuracy.
- ❖ Comparison: This paper focuses on a success classification task, similar to our Profit/Loss classification task. Our CatBoost model achieved an overall accuracy of 92% for Profit/Loss classification. This result shows that the performance of our CatBoost model significantly exceeds the accuracy reported for the best model in this study

(86% for Neural Network), indicating the effectiveness of our project's approach for the binary classification task.

2. "Movie Revenue Prediction Using Machine Learning Models" (Udandaraao & Gupta)

- ❖ Paper Goal: Predict movie revenues.
- ❖ Models and their Performance: Gradient Boosting (0.8242), XGBoosting (0.8102), and Random Forest (0.7786) - likely R^2 scores.
- ❖ Comparison: This study closely resembles our revenue prediction task. Our Random Forest model achieved an R^2 of 0.7761. In comparison, Random Forest in this study achieved an R^2 of 0.7786, showing a significant convergence in performance between the two models for the same type. However, the Boosting models in this study (Gradient Boosting, XGBoost) outperformed our Random Forest. This indicates that there is room for improving the performance of our regression models, perhaps through hyperparameter tuning or exploring additional features.

3. "Long-Range Movie Box Office Prediction Based on Machine Learning" (Xu)

- ❖ Paper Goal: Predict long-term box office revenue before a film's release.
- ❖ Models and their Performance: Random Forest (0.738), linear regression (0.608), and neural network (0.625) - likely R^2 scores.
- ❖ Comparison: This study focuses on revenue prediction, resembling our task in this regard. Our Random Forest model (0.7761) outperforms the performance of Random Forest in this study (0.738). This suggests that the rich set of features and data preprocessing process in our project contributed to achieving better predictive performance for movie revenues compared to this study.

4. "Predicting Movie Box Office Based on Machine Learning, Deep Learning, and Statistical Methods" (Zheng)

- ❖ Paper Goal: Predict global box office revenue.
- ❖ Models and their Performance: Bidirectional LSTM (0.7364), XGBoost (0.7084), Random Forest (0.6724) - R^2 scores.
- ❖ Comparison: This study also deals with revenue prediction and uses machine learning and deep learning models. Compared to our best regression model (Random Forest 0.7761), the performance of Random Forest and XGBoost in this paper (0.6724 and 0.7084) is lower than our performance. However, the deep learning model Bidirectional LSTM in this paper achieved an R^2 of 0.7364, which approaches the performance of our Random Forest. This highlights that deep learning models may offer robust alternatives, especially when dealing with complex data.

c. Conclusions from Comparison

This comparison shows that the work accomplished in this project achieves competitive performance, and in some cases, surpasses the results reported in previous research, particularly in the film profitability classification task, where the CatBoost model achieved very high accuracy. In the revenue prediction task, the Random Forest model showed strong performance, exceeding some referenced research and approaching the best results reported in other studies using similar models.

However, our comparison with research that used boosting models (like Gradient Boosting and XGBoost) or deep learning (like LSTM) indicates that there is always room for further improvement. Future work could involve deeper exploration of hyperparameter tuning for these models, integrating additional features, or experimenting with more complex deep learning architectures that might exceed current performance.

VIII. Conclusion and Future Work

This project represents a comprehensive effort to predict movie revenue and profitability, successfully integrating and processing multiple complex datasets as a strong foundation for modeling. Experiments demonstrated the effectiveness of the Random Forest model in predicting revenue, achieving competitive performance even after addressing the challenge of zero values with logarithmic transformations. The CatBoost model also excelled in the film profitability classification task, achieving high accuracy exceeding 90%, demonstrating its superior ability to handle class imbalance. Overall, these results place the project in a competitive position compared to previous research in the field. For future work, it is suggested to improve the accuracy of regression models through hyperparameter tuning and exploring deep learning models, in addition to developing more advanced methods to enhance model interpretability for a deeper understanding of the factors influencing movie success.

IX. References

1. Long-Range Movie Box Office Prediction Based on Machine Learning, Highlights in Science, Engineering and Technology, vol. 92, pp. 309–310, 2024. [Online]. Available: <https://drpress.org/ojs/index.php/HSET/article/view/19900/19480>
2. V. Udandaraao and P. Gupta, "Movie Revenue Prediction using Machine Learning Models," arXiv preprint arXiv:2405.11651, May 2024. [Online]. Available: <https://arxiv.org/abs/2405.11651>
3. M. Agarwal, S. Venugopal, R. Kashyap, and R. Bharathi, "A Comprehensive Study on Various Statistical Techniques for Prediction of Movie Success," arXiv preprint arXiv:2112.00395, Dec. 2021. [Online]. Available: <https://arxiv.org/abs/2112.00395>
4. Y. Zheng, "Predicting Movie Box Office Based on Machine Learning, Deep Learning, and Statistical Methods," Applied and Computational Engineering, vol. 94, pp. 20–32, 2024. [Online]. Available: <https://ewadirect.com/proceedings/ACE/article/view/16032>
5. A. Pawar, "Movies Daily Update Dataset," Kaggle. [Online]. Available: <https://www.kaggle.com/datasets/akshaypawar7/millions-of-movies>. [Accessed: Jul. 15, 2025].
6. IMDb, "IMDb Non-Commercial Datasets," updated Mar. 18, 2024. [Online]. Available: <https://developer.imdb.com/non-commercial-datasets/>. [Accessed: Jul. 15, 2025].
7. unaniamad, "The Oscar Award, 1927 - 2025," Kaggle, updated Apr. 2025. [Online]. Available: <https://www.kaggle.com/datasets/unaniamad/the-oscar-award>. [Accessed: Jul. 15, 2025].
8. Ö. F. Eker, "Forbes Celebrity 100 since 2005," Kaggle, updated Jun. 9, 2020. [Online]. Available: <https://www.kaggle.com/datasets/slayomer/forbes-celebrity-100-since-2005>. [Accessed: Jul. 15, 2025].